Inverse Methods [EAS 6134]
Problem Set #3

C. Michael Haynes

1) Exercise 1)a-i in ch2

a) Since the errors of measurement are independent and normally distributed with a mean of 0, we proceed with a least-squares approach.

The data and measurement error along with the least-squares regression fit are included in the top panel. Measurement residuals are then depicted on a color-coded shading in the bottom panel.

b) For a given model covariance $\mathrm{cov}(\vec{m})$, the correlation matrix is given by

$$\mathrm{Corr}(\vec{m}) = \frac{\mathrm{Cov}(\vec{m})}{\sqrt{\mathrm{Var}(m_i)\,\mathrm{Var}(m_j)}} .$$

→ The diagonals are unity, since each variable is auto-correlated.

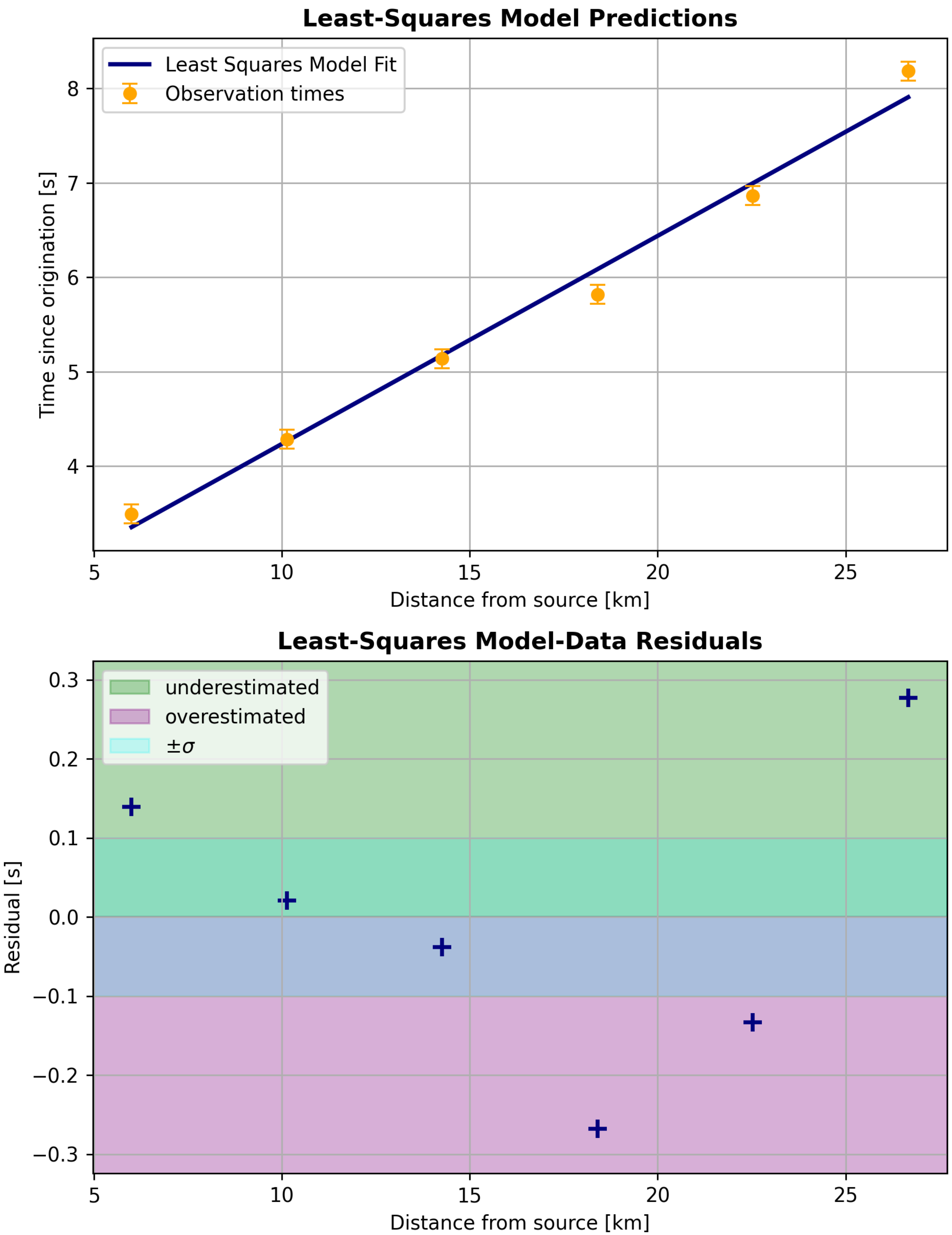
→ the off-diagonals are near -1, implying that the error ellipse is highly elongated: both columns (rows) of $\mathrm{corr}(\vec{m})$ are inclined by $\approx 45°$ from either "axis": (t_0, s_2).

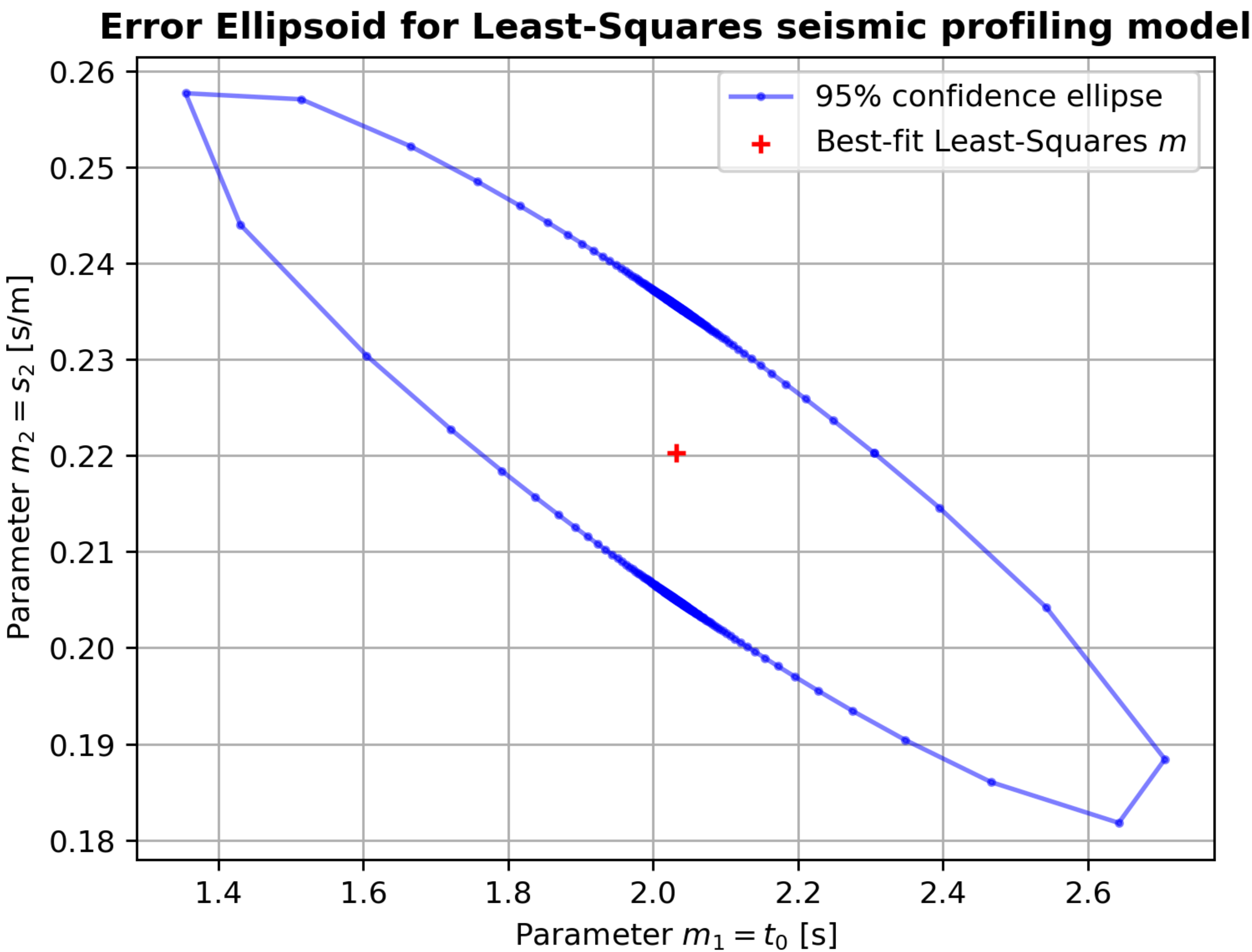
→ This indicates a small error in one component of the 2-parameter model $\vec{m} = (t_0, s_2)$, and a large error in the other.

c) The error ellipse is included in an attached figure. We sample it at $n = 300$ points to actually resolve it near the major axis.

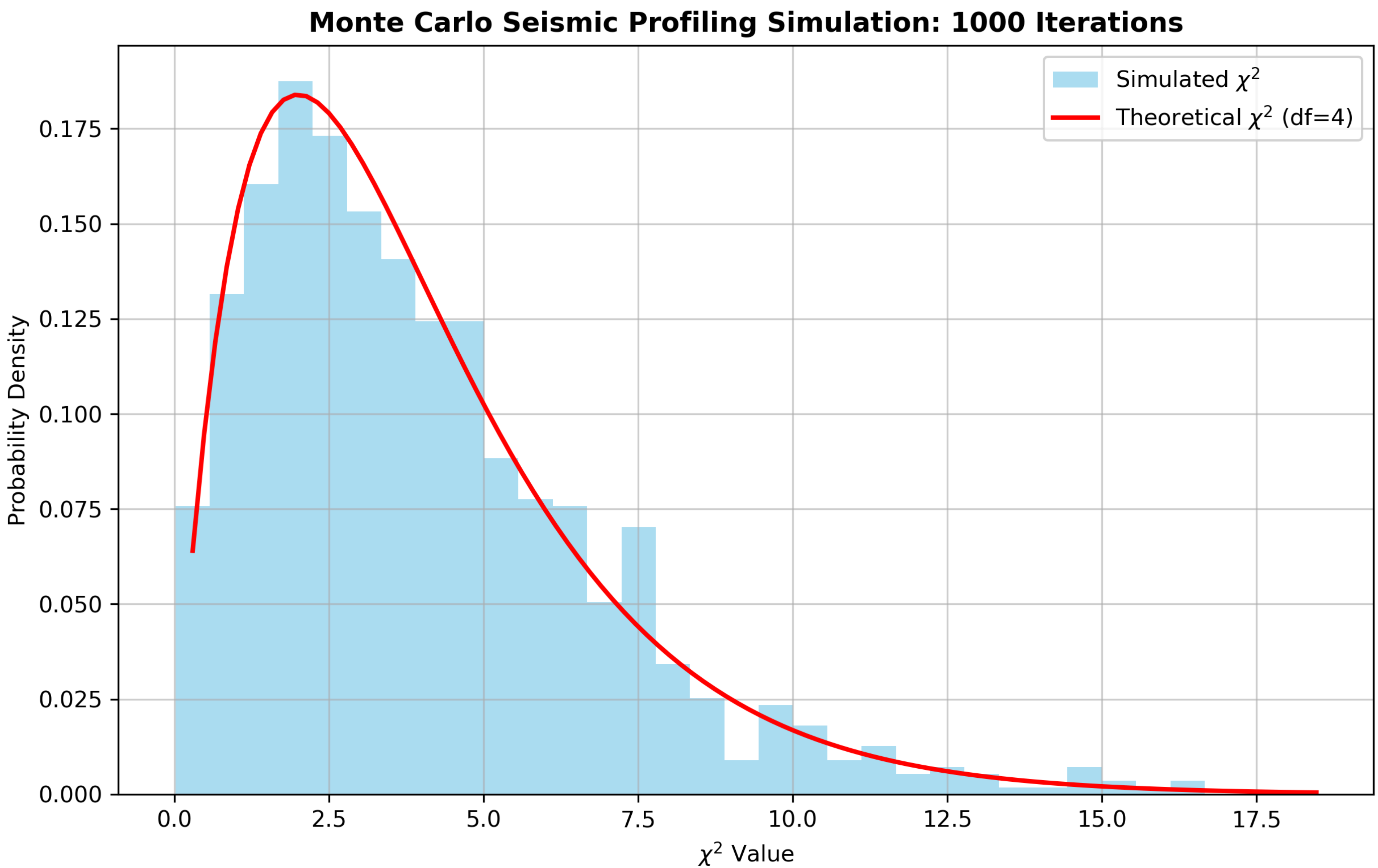
d) p-value: $8.7916 \cdot 10^{-4}$.

e) The curve and associated histogram showing these results is attached in a figure.


Least-Squares Model Predictions
Least Squares Model Fit
Observation times
Time since origination [s]
Distance from source [km]
Least-Squares Model-Data Residuals
underestimated
overestimated
±σ
Residual [s]
Distance from source [km]



Error Ellipsoid for Least-Squares seismic profiling model
95% confidence ellipse
Best-fit Least-Squares m
0.26
0.25
0.24
0.23
0.22
0.21
0.20
0.19
0.18
1.4
1.6
1.8
2.0
2.2
2.4
2.6
Parameter $m_2 = s_2$ [s/m]
Parameter $m_1 = t_0$ [s]



Monte Carlo Seismic Profiling Simulation: 1000 Iterations
Simulated χ^2
Theoretical χ^2 (df=4)
Probability Density
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
χ^2 Value

f) our p-value, while not large at all, is still a higher value than it may be with a truly "wrong" model which would produce several orders of magnitude smaller. It is hence roughly consistent. The histogram is extremely consistent with the calculated distribution. The reason the p-value is small may be due to either (a) the data represent an unlikely realization, or (b) the data errors are underestimated or not precisely normally-distributed. We assume that the model is appropriate, but a systemic misrepresentation of the forward model would also result in a very low p-value, and we certainly don't expect that our model encodes the complete seismic physics.

g) See attached figure.

h) 95% confidence interval: $\Delta^2 = F^{-1}_{\chi^2, n=4}(0.95) \approx 9.48$

$$t_0 : (1.3461, 2.7186) \rightarrow \text{less constrained}$$

$$s_2 : (0.1817, 0.2589) \rightarrow \text{highly constrained}$$

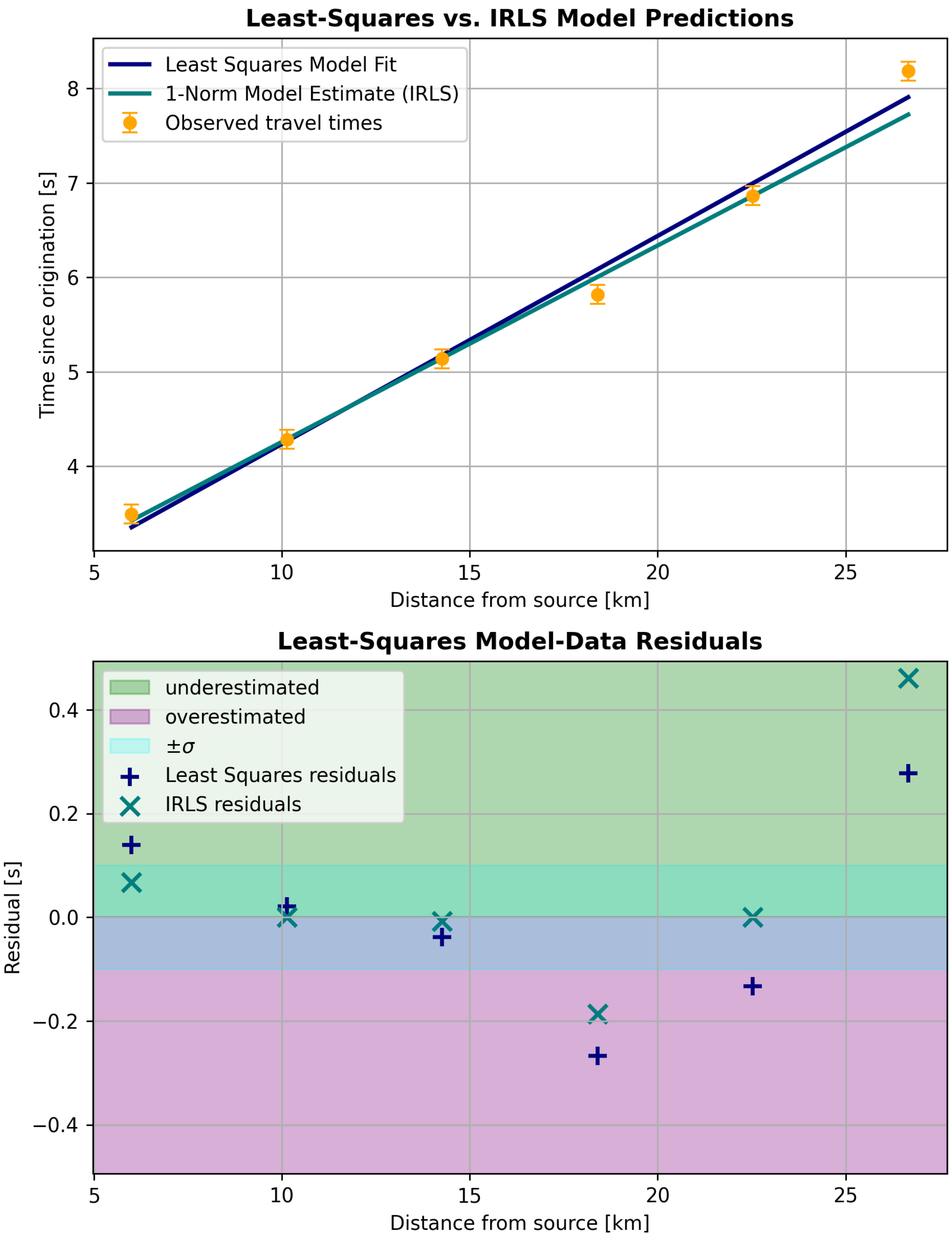
i) The bulk of the misfit is carried by the 'last' data point (25 km measurement), and the IRLS 'prioritizes' this point less than the others to recover a stronger overall fit.

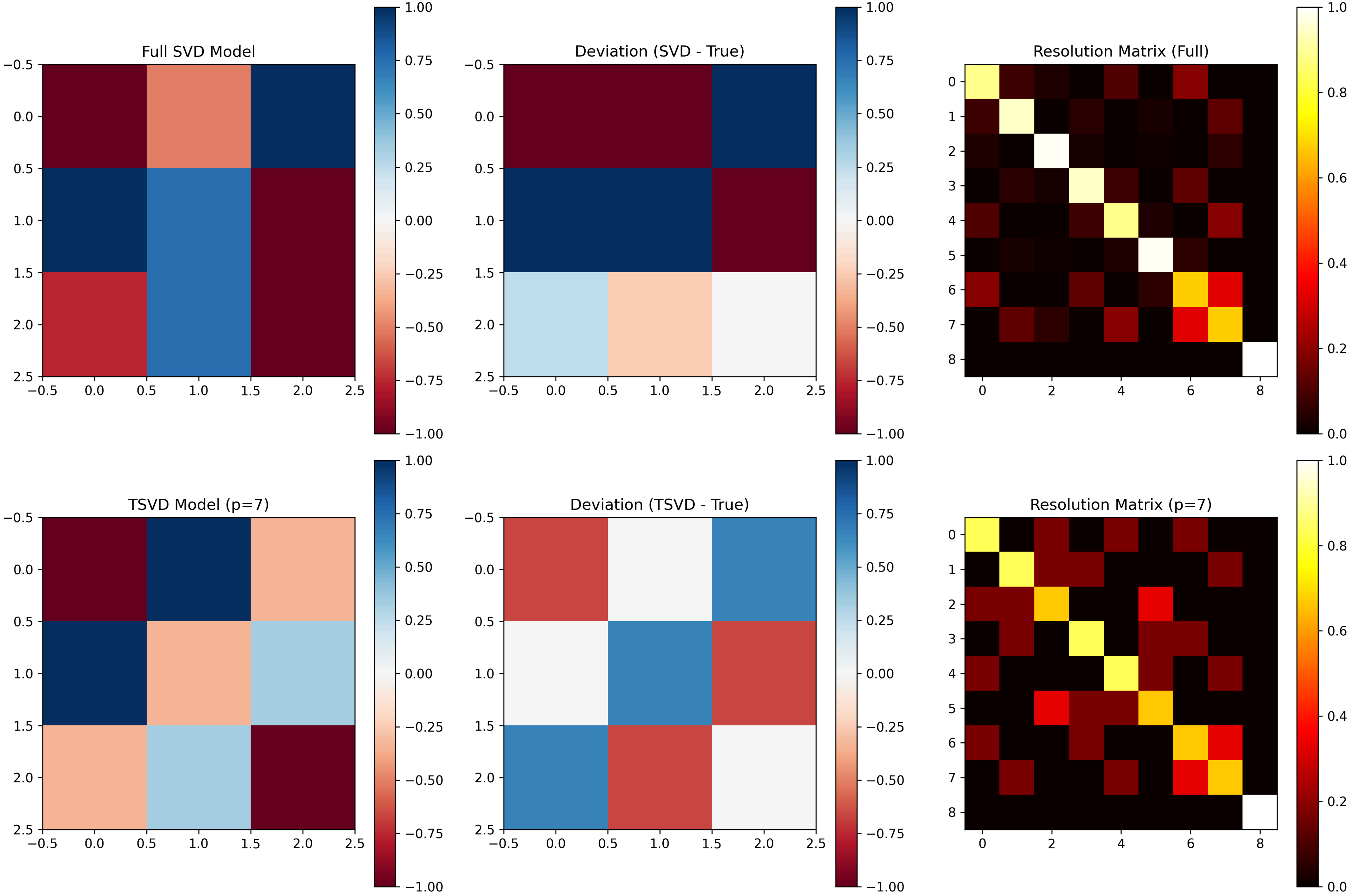
Exercise 2 in ch. 3

→ yes, 3 block vals are recovered exactly, using truncated SVD. Evaluating the resolution matrix,

$$\underline{\underline{R}} \equiv \underline{\underline{V}}_P \cdot \underline{\underline{V}}_P^T,$$

shows that the trend in model recovery is illustrated/mirrored by the morphology of the resolution matrix. Put simply, high resolution along the diagonal and at two off-diagonal locations allow to recover three values exactly. The influence of truncation is evaluated by comparing the truncated solution with the generic one. See attached figure.


Least-Squares vs. IRLS Model Predictions
Least Squares Model Fit
1-Norm Model Estimate (IRLS)
Observed travel times
Time since origination [s]
Distance from source [km]
Least-Squares Model-Data Residuals
underestimated
overestimated
±σ
Least Squares residuals
IRLS residuals
Residual [s]
Distance from source [km]


Full SVD Model
Deviation (SVD - True)
Resolution Matrix (Full)
TSVD Model (p=7)
Deviation (TSVD - True)
Resolution Matrix (p=7)

3) Exercise 4.2 (a)-(f) in Ch. 3

a) Our model representation for the seismic slowness perturbations $\underline{\underline{G}} \cdot \vec{m} = \vec{d}$ admit an operator $\underline{\underline{G}}$ with shape (94, 256) where $n_{scans} = 94$, and $n_{elements} = 256$. The elements of the model are on a 16×16 grid of discretized sandstone quarry (cuboid-shaped).

The rank$(\underline{\underline{G}}) = 90$, indicating that $\underline{\underline{G}}$ is not full rank, and so the model null space is non-empty

b) Because rank$(\underline{\underline{G}}) = 90$, the null space has a rank of 4. Hence, the model null space N is spanned by the final 4 columns of $\underline{\underline{V}}$, where SVD yields

$$\underline{\underline{G}} = \underline{\underline{U}} \cdot \underline{\underline{S}} \cdot \underline{\underline{V}}^T .$$

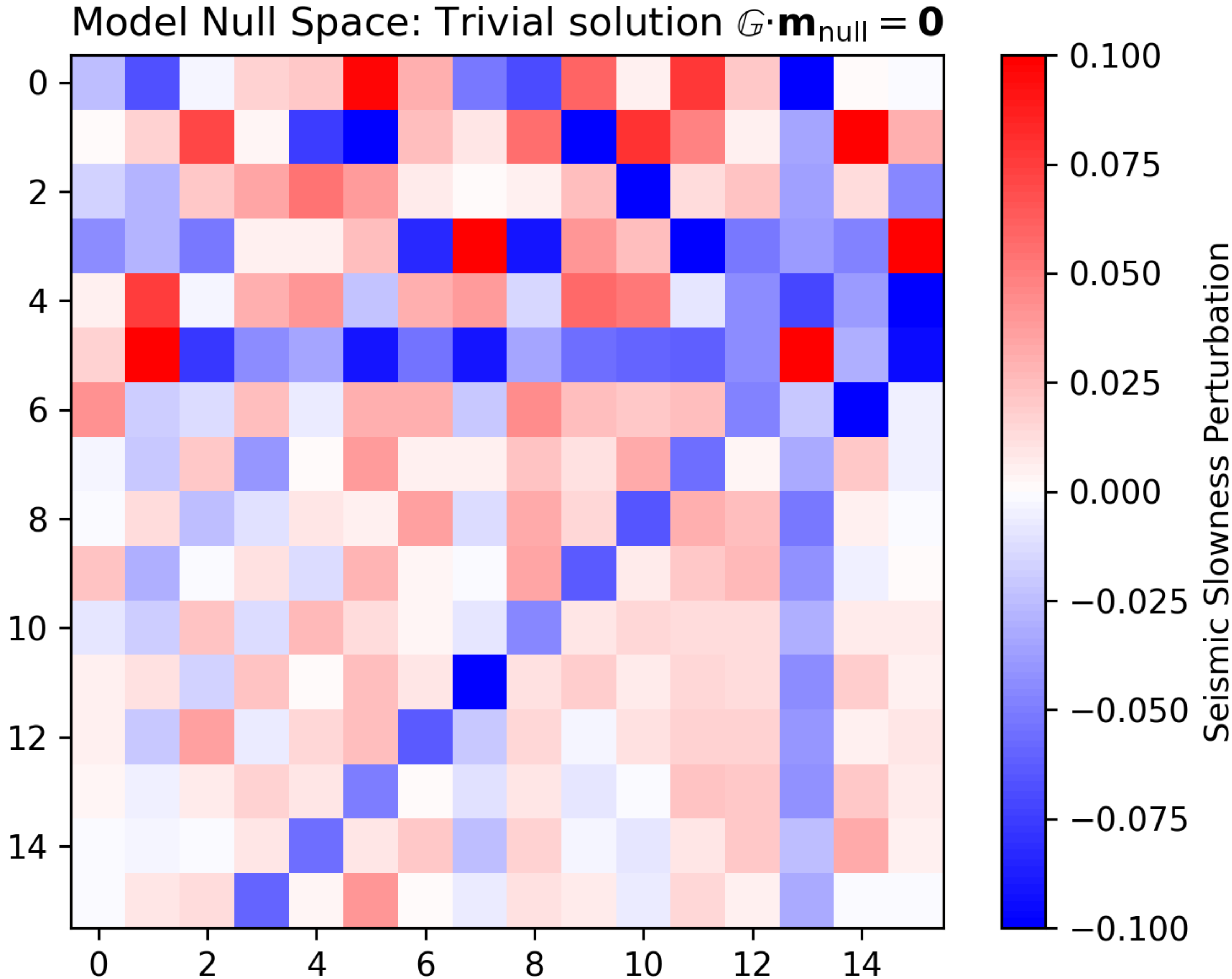
In other words, there are 90 singular values that are nonzero, so SVD can be written as

$$\underline{\underline{G}} = \begin{bmatrix} \underline{\underline{U}}_p & \underline{\underline{U}}_0 \end{bmatrix} \cdot \begin{bmatrix} \underline{\underline{S}}_p & \\ & \underline{\underline{0}} \end{bmatrix} \cdot \begin{bmatrix} \underline{\underline{V}}_p^T \\ \underline{\underline{V}}_0^T \end{bmatrix} ,$$

where $\underline{\underline{V}}_0$ represents the null space of $\underline{\underline{G}}$. We depict some of ~~the~~ (flattened) vectors that represent a basis to the null space of the model, N. Using these, we can construct a solution to the trivial problem

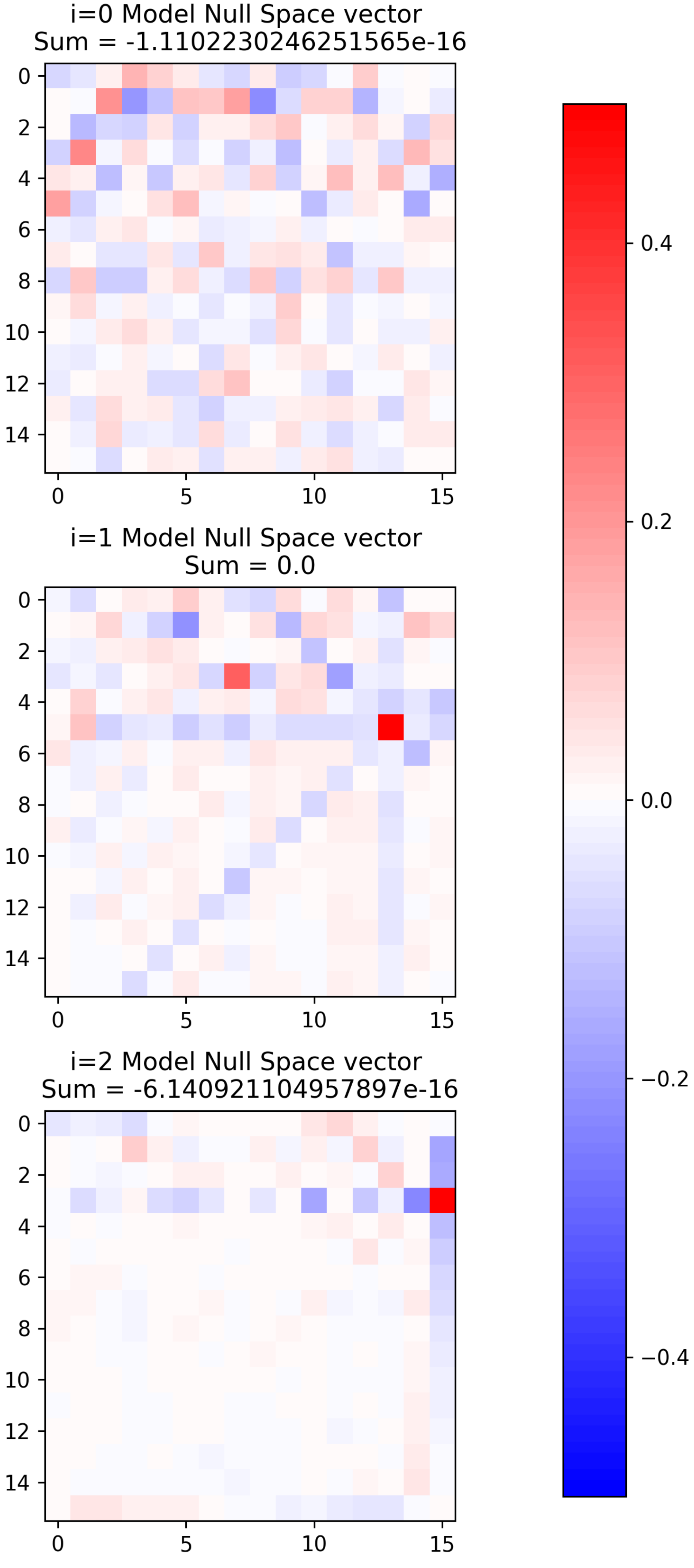
$$\underline{\underline{G}} \cdot \vec{m}_{null} = \vec{0} .$$

$\vec{m}_{null}$ is also depicted in a flattened shape. The null space basis also indicates where velocity perturbations have little influence on the model we recover.



Model Null Space: Trivial solution $\mathbb{G}\cdot\mathbf{m}_{\mathrm{null}} = \mathbf{0}$
0
2
4
6
8
10
12
14
0.100
0.075
0.050
0.025
0.000
−0.025
−0.050
−0.075
−0.100
Seismic Slowness Perturbation

Model Null Space: Reshaped Visualization (256 --> 16 x 16)



c) The resolution Matrix $\underline{\underline{V}}_p \cdot \underline{\underline{V}}_p^T$ indicates that there are 4 instances of perfect resolution, likely the 'corners' of the block where the model is highly constrained. The 256×256 resolution matrix is included both using linear & log scales, since many off-diagonal elements have a very low associated resolution.

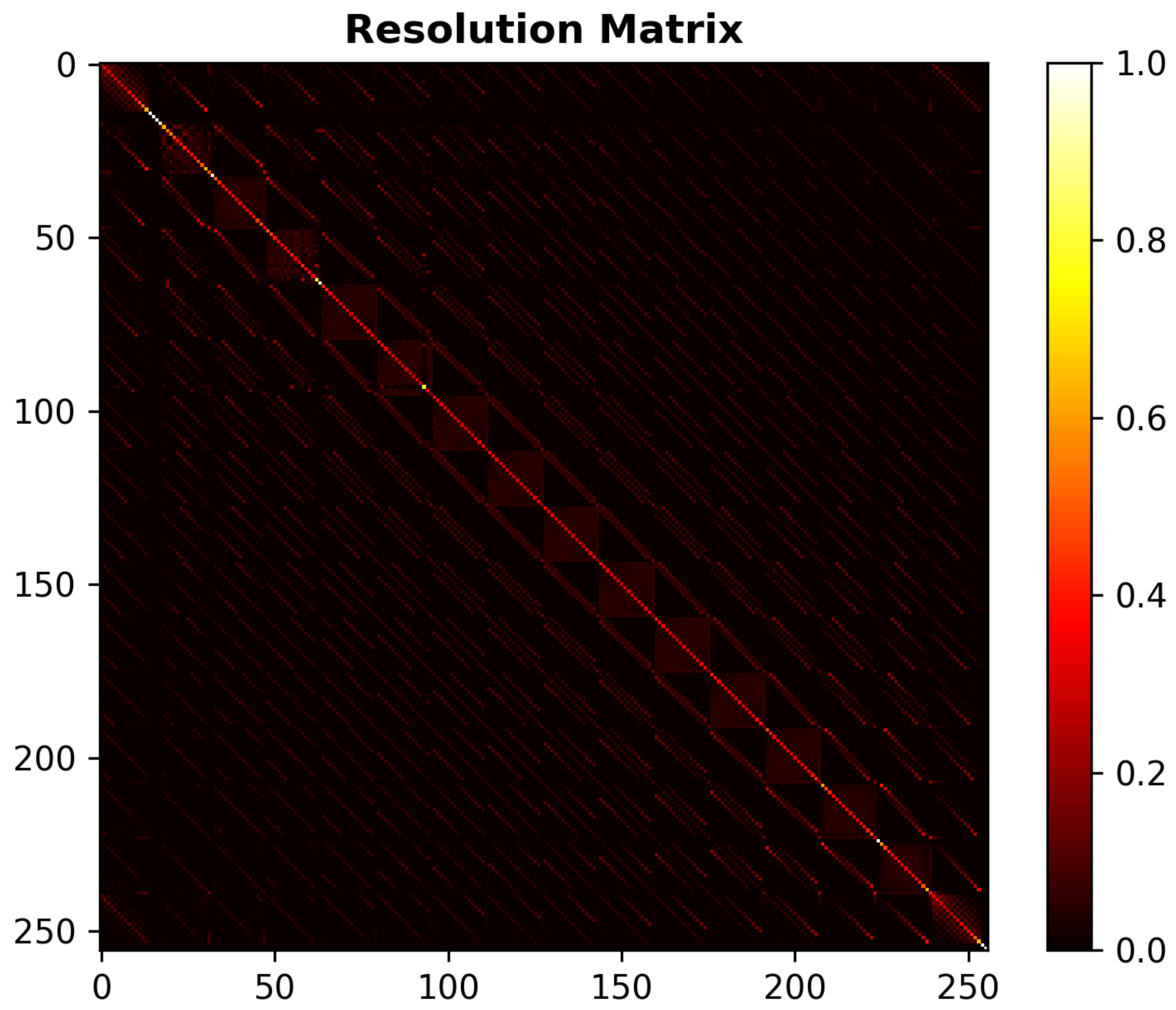
d) We use truncated SVD (with a threshold of 10^{-15}) to produce the pseudo-inverse solution

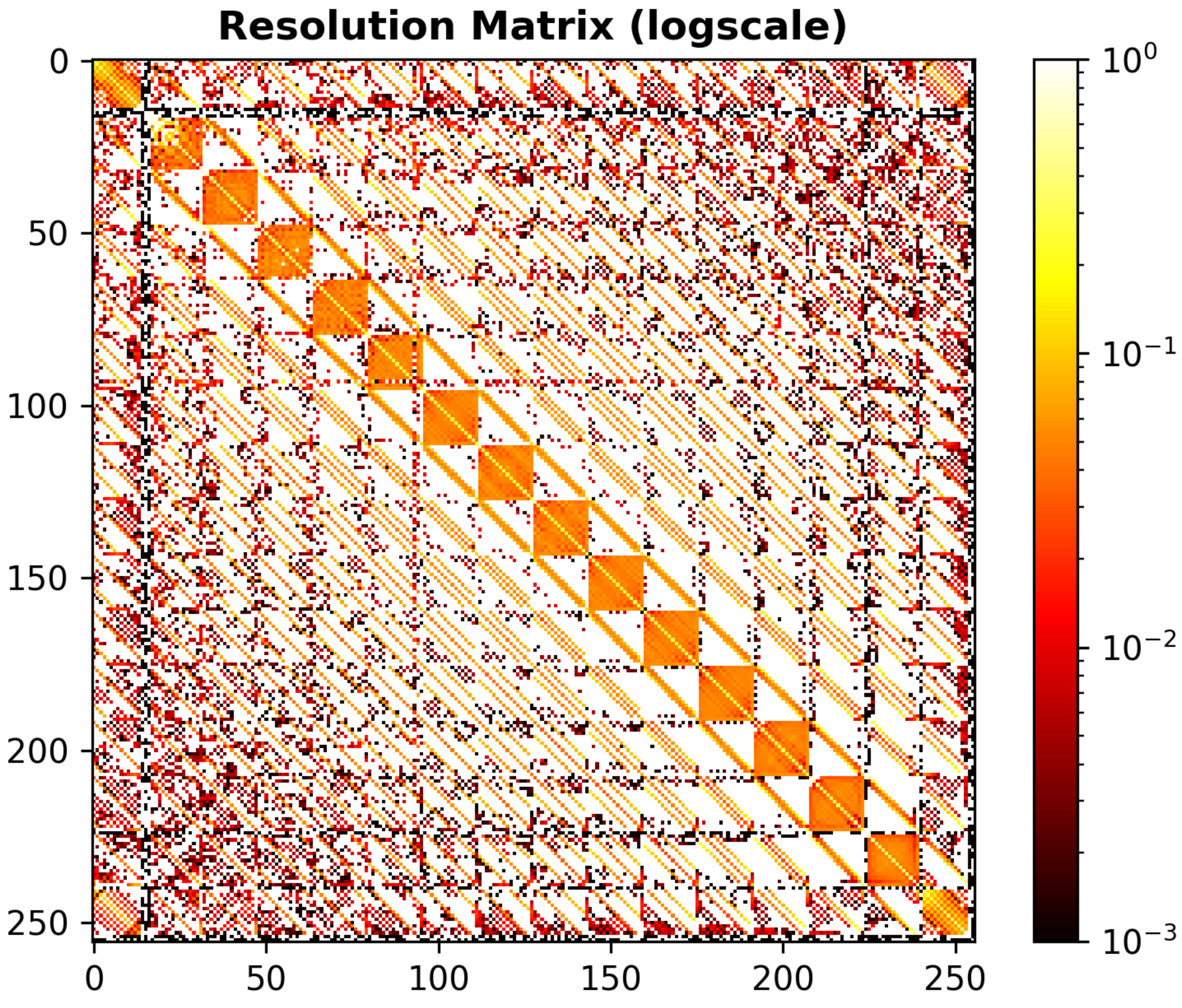
$$\vec{m}_+ = \underline{\underline{G}}_+ \cdot \vec{d} \qquad \left(\underline{\underline{G}}_+ = \underline{\underline{V}}_p \cdot \underline{\underline{S}}_p^{-1} \cdot \underline{\underline{U}}_p^T \right)$$

We notice that there appear to be three main morphological features associated with the recovered slowness model (depicted on a colormap where red ⇒ lower velocity and blue ⇒ higher velocity / lower slowness):

- → two main blocks (centered at (6,6) & (12,12)) of reduced slowness, with perturbations nearing 30-50% the value of the "background" material slowness. These blocks are where the fossils are most likely located.
- → A reduced slowness (minor, 10-20%) along the <u>entire</u> diagonal trace. This is likely due to lack of perfect constraint along the entire main diagonal, so some reduced slowness "creeps out" of the primary blocks where the velocity perturbations are largest.
- → A 'diffuse' background of velocity perturbations only on the order of 5-10% the nominal slowness value $s_0 = 1/3000$ [s/m]. These are present off the main diagonal, and likely stem from <u>both</u> imperfect scan coverage ($\underline{\underline{G}}$ is not full rank) <u>and</u> the systematic noise (Gaussian, centered around the measurements) present in the observed travel times.

<u>Max perturbation: $+4.24 \cdot 10^{-5}$ [s/m]</u>   <u>Min perturbation: $-9.87 \cdot 10^{-5}$ [s/m]</u>


Resolution Matrix
0
50
100
150
200
250
0
50
100
150
200
250
1.0
0.8
0.6
0.4
0.2
0.0



Resolution Matrix (logscale)
0
50
100
150
200
250
0
50
100
150
200
250
10^0
10^{-1}
10^{-2}
10^{-3}

e) The diagonals of the resolution matrix show that the maximum resolution is acheived at the corners of the block. The resolution is inherently the weakest near the center of the block, where the slowness values are the least constrained due to the selected array of ray paths that were considered. This is because each of the constraint equations/ray paths associated with the middle group of block cells pass through ~16 other cells, distributing the uncertainty.

We note that the offset from the upper left and right edge are likely caused by an artifact due to the truncation of $\underline{\underline{V}}_p$. This may be unphysical.

f) Since we have used the basis vectors of the null space to produce an arbitrary trivial solution $\vec{M}_{null}$, we have that

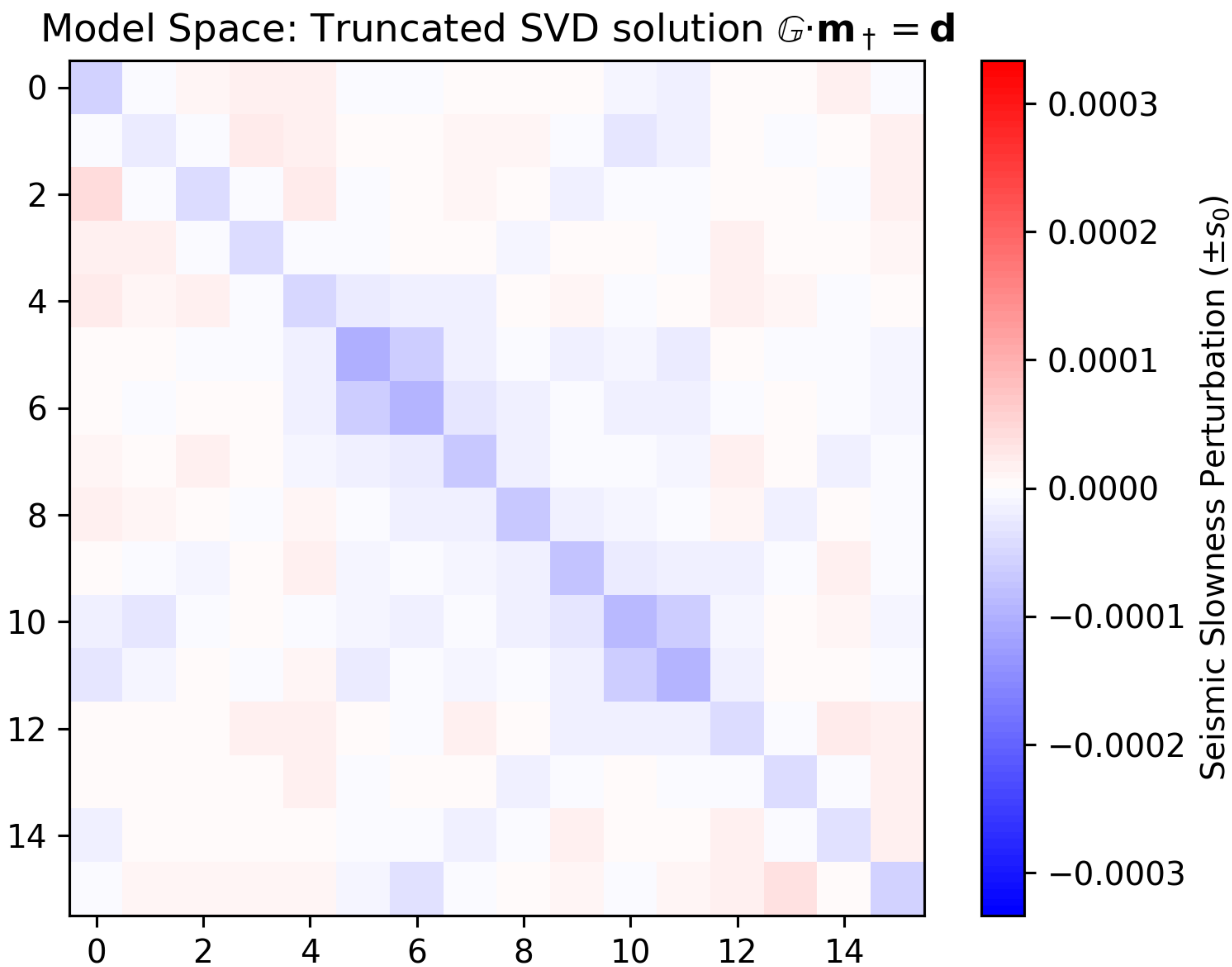
$$\vec{M}_+^{\# \varepsilon \text{(ignore)}} = \underline{\underline{G}}_+^{\#} \cdot \vec{d} \quad \text{and} \quad \underline{\underline{G}} \cdot \vec{M}_{null} = \vec{0}. \quad \text{Then,}$$

$$\underline{\underline{G}} \cdot \vec{M}_+ = \vec{d}, \quad \text{so} \quad \underline{\underline{G}} \cdot \vec{M}_+ + \underline{\underline{G}} \cdot \vec{M}_{null} = \vec{d} + \vec{0}$$

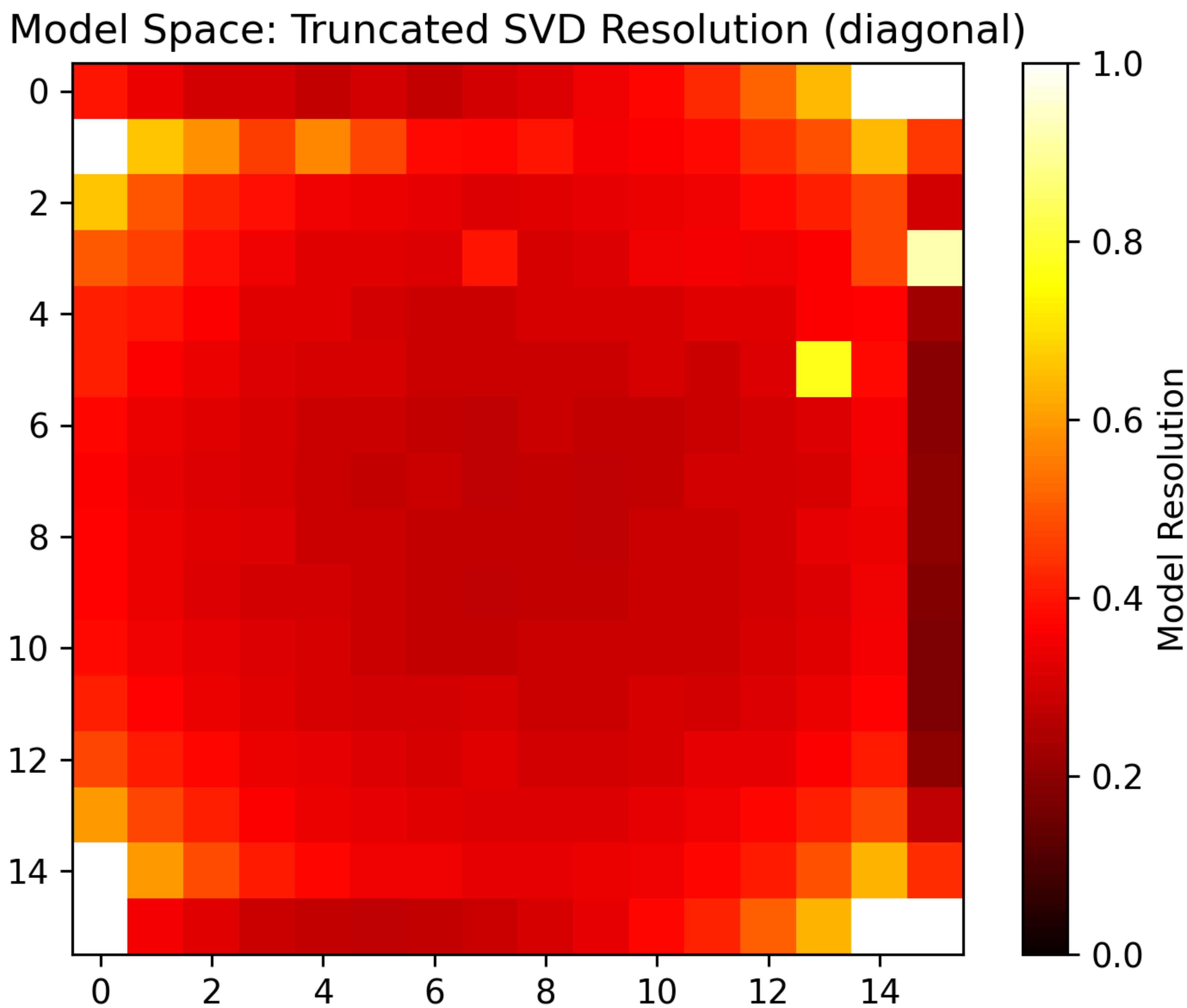
$$\Rightarrow \quad \underline{\underline{G}} \cdot (\vec{M}_+ + \vec{M}_{null}) = \vec{d}$$

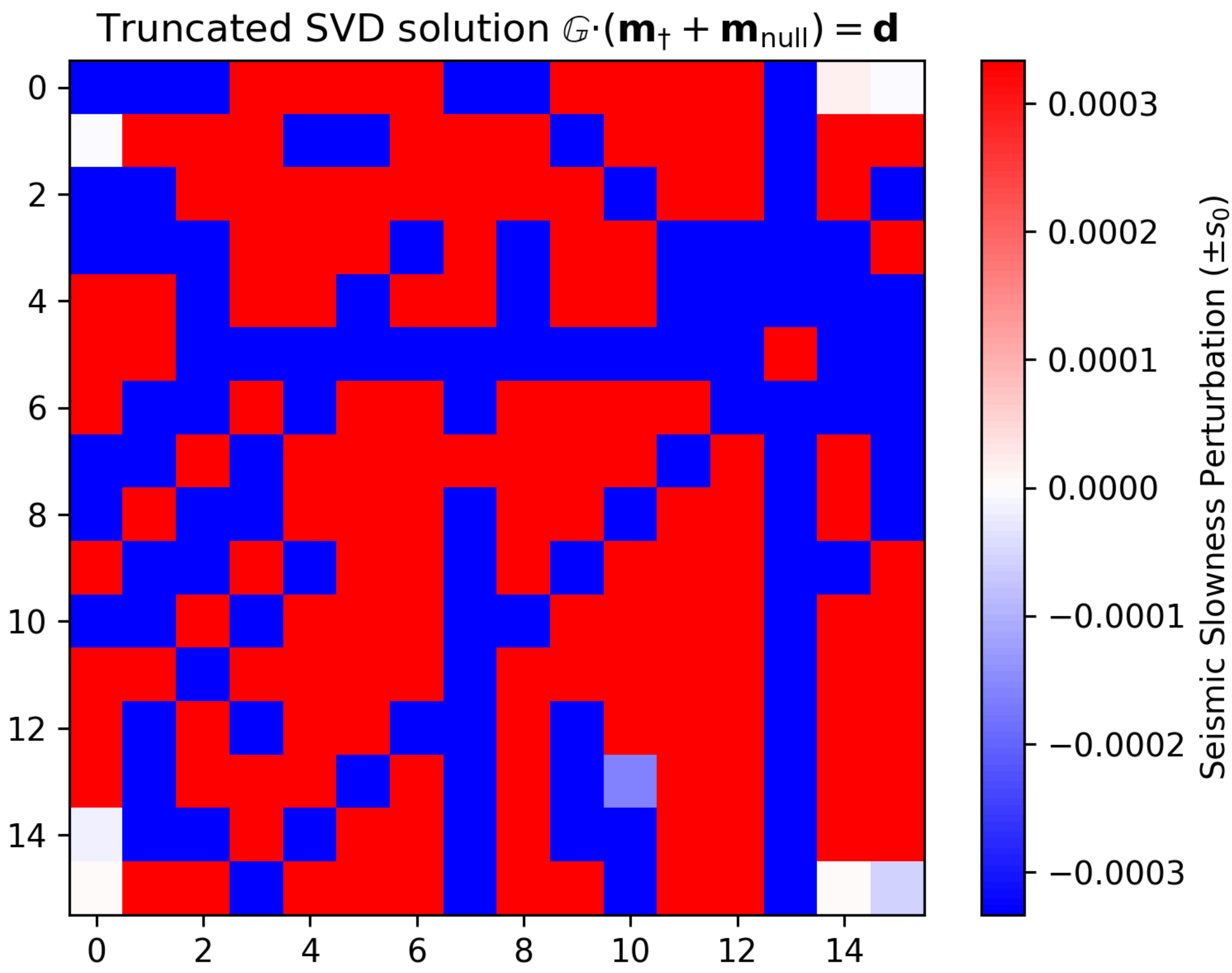
Hence, like $\vec{M}_+$, $\vec{M}_+ + \vec{M}_{null}$ is <u>also</u> a solution for all $\vec{M}_{null} \in N$.

We use our $\vec{M}_{null}$ from part (b) to demonstrate such a solution to the system. Please see the attached figures depicting such a "wild" model.



Model Space: Truncated SVD solution $\mathbb{G}\cdot\mathbf{m}_\dagger = \mathbf{d}$
Seismic Slowness Perturbation ($\pm s_0$)
0.0003
0.0002
0.0001
0.0000
−0.0001
−0.0002
−0.0003
0
2
4
6
8
10
12
14


Model Space: Truncated SVD Resolution (diagonal)
0
2
4
6
8
10
12
14
Model Resolution
0.0
0.2
0.4
0.6
0.8
1.0


Truncated SVD solution $\mathbb{G}\cdot(\mathbf{m}_\dagger + \mathbf{m}_{\mathrm{null}}) = \mathbf{d}$
0
2
4
6
8
10
12
14
0.0003
0.0002
0.0001
0.0000
−0.0001
−0.0002
−0.0003
Seismic Slowness Perturbation ($\pm s_0$)

5) Exercise 5 in ch. 5

a) At each point of observation y_i, the corresponding integral of the kernel is

$$G_{ij} = \int_{x_j}^{x_{j+1}} x e^{-y_i x} \, dx$$

Integration by parts yields

$$\int x e^{-yx} dx = -\frac{1}{y}\left(x e^{-yx} + \frac{1}{y} e^{-yx} \right)$$

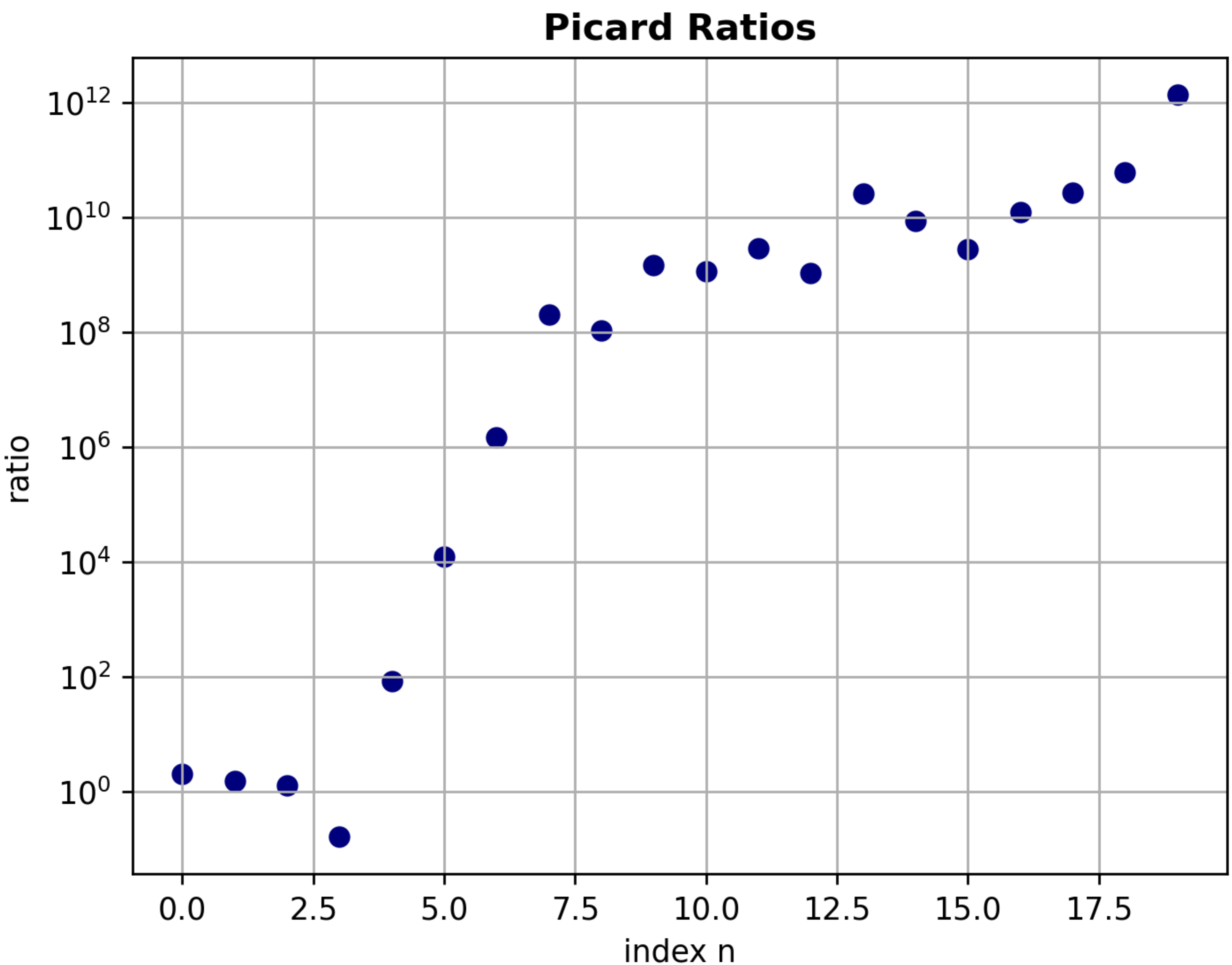
$$\Rightarrow G_{ij} = -\frac{1}{y_i}\left(\left(\frac{x_{j+1} e^{-y_i x_{j+1}}}{1} + \frac{e^{-y_i x_{j+1}}}{y_i} \right) - \left(x_j e^{-y_i x_j} + \frac{e^{-y_i x_j}}{y_i} \right) \right)$$

The solution using M-P inversion yields model values on the order of 10^{11}

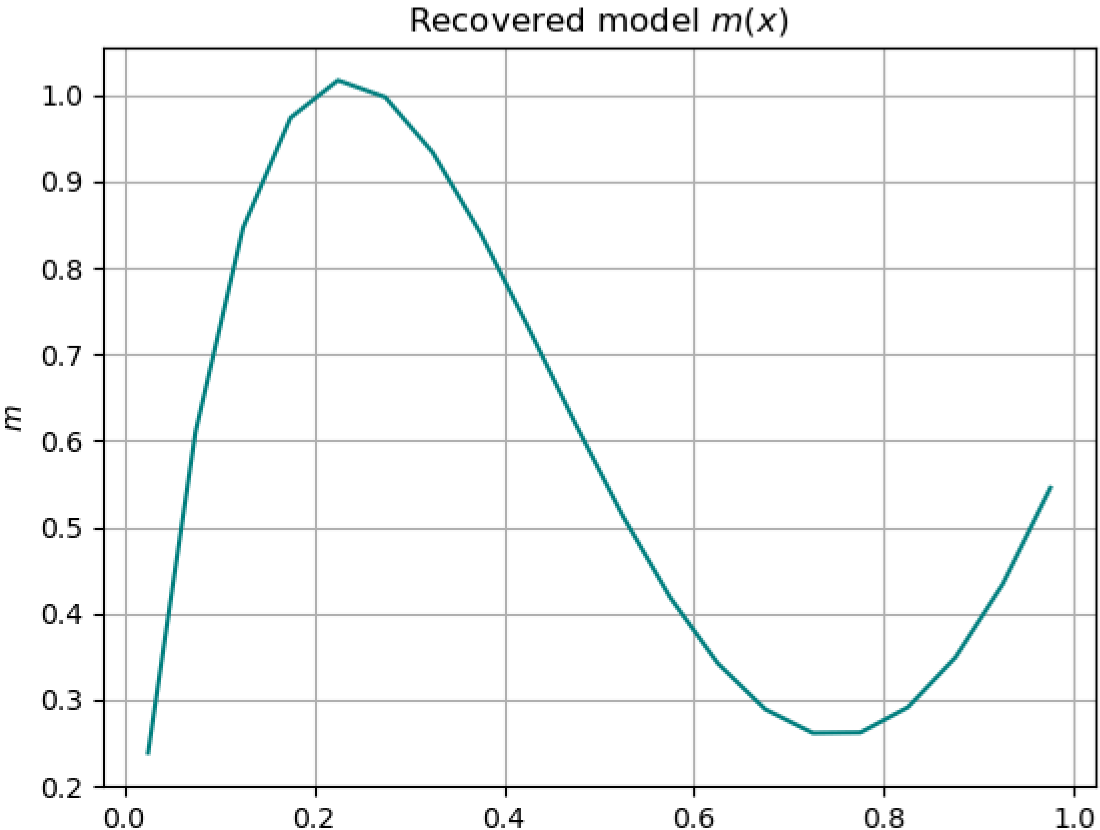
b) The condition number of $\underline{\underline{G}}$ is very large, on the order of 10^{15}: $(2.25 \cdot 10^{15}) = \text{cond}\,\underline{\underline{G}}$. The result of such large solution values, as indicated by the condition numbers, is amplified noise. Since the measurements only contain 4 digits of accuracy, the M-P inverse $\underline{\underline{G}}_+$ produces unfiltered noise amplification.

c) By performing truncated SVD where we keep only the first 4 indices, motivated by the Picard ratio behavior (see uptick after $n=4$) in the attached figure, we recover a model solution that has far less noise and yields values on the order of the inputs and outputs to the model, rendering confidence. ~~The condition number of the new, truncated~~ ~~G_+ is~~ [not asked for I realized] ~~$\text{cond}(\underline{\underline{G}}_+) =$~~

We include a figure depicting $M(x)$.



Picard Ratios
10^{12}
10^{10}
10^{8}
10^{6}
10^{4}
10^{2}
10^{0}
ratio
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
index n


Recovered model $m(x)$
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
m
0.0
0.2
0.4
0.6
0.8
1.0
y